CLASSIFICATION OF INDIAN COW BREEDS USING MACHINE LEARNING

A Minor Project Dissertation Submitted
in Partial Fulfilment of the Requirements
for the Degree of

MINOR DEGREE

in **Data Science**

by

Nupur Dipakbhai Upadhyay
(Reg. No. IMS22191)



to

SCHOOL OF DATA SCIENCE
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INDIA - 695 551

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CERTIFICATE

This is to certify that this dissertation entitled **"Classification of Indian Cow breeds using Machine Learning"** submitted by **Nupur Dipakbhai Upadhyay (Reg. No.: IMS22191)** towards the partial requirement of **Minor Degree** in **Data Science**, has been duly examined by the thesis committee appointed by the institute. The committee deems the candidate's work satisfactory and recommends that the report be accepted.





Dr Saptarshi Bej	**Dr N. Sadananda Singh**
Supervisor	Co-Supervisor

5 May 2026
Thiruvananthapuram - 695 551

DECLARATION

I, **Nupur Dipakbhai Upadhyay (Reg. No.: IMS22191)**, hereby declare that this dissertation entitled **"Classification of Indian Cow breeds using Machine Learning"**, submitted to Indian Institute of Science Education and Research, Thiruvananthapuram, towards the partial requirement of **Minor Degree** in **Data Science**, is a bona fide record of original work carried out by me under the supervision of **Dr Saptarshi Bej** and **Dr N. Sadananda Singh**.

This dissertation has never been submitted in part or in full for a degree, diploma or fellowship to this or any other university before. I duly acknowledge all external contributions, statements, datasets or results used, and have listed their sources with adequate detail in the bibliography.



.....

Nupur Dipakbhai Upadhyay

Reg. No.: IMS22191

In my capacity as the project supervisor for the aforementioned candidate, I confirm that the above statements by the candidate are true to the best of my knowledge.



.....

Dr Saptarshi Bej

Supervisor



.....

Dr N. Sadananda Singh

Co-Supervisor

5 May 2026
Thiruvananthapuram - 695 551

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Nupur Dipakbhai Upadhyay

ABSTRACT

Student Name: **Nupur Dipakbhai Upadhyay** Reg. No.: **IMS22191**
Degree: **Minor Degree** Dept.: **School of Data Science**
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India possesses diverse indigenous cattle breeds with distinct genetic architectures linked to important traits such as disease resistance and milk production. However, traditional breed identification methods based on phenotypes or limited genetic markers are often unreliable or costly. This study explores the use of machine learning for accurate breed classification using whole-genome sequencing (WGS) and SNP genotype data, while also aiming to reduce the number of required genetic markers.

A dataset of 344 samples across 9 breeds was analyzed using 11,118 SNPs obtained after quality control and filtering. Multiple supervised models were evaluated, with logistic regression using L1 regularization emerging as the best-performing model after hyperparameter tuning and 10-fold cross-validation. The model achieved high classification performance, with an AUC of 0.999.

Despite limitations such as small sample size and class imbalance, this approach demonstrates the potential for cost-effective, high-accuracy genomic breed classification in Indian cattle.

Keywords: ML, Indian Cow, Classification, Genomics

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Chapter 1

Introduction

Cattle are often regarded as the cornerstone of agriculture. From providing milk and meat to serving as draught animals, they have been indispensable companions to farmers worldwide[1]. In India, cattle hold significant socio-economic value, deeply intertwined with the nation's religion, culture, and economy. India is the world's largest producer and consumer of milk, with national production reaching 239.30 million tonnes in 2023–24 and per capita availability of 471 g/day[2]. Cattle are particularly vital in rural livelihoods, where marginal and small farmers, along with landless labourers, own about 70% of the total livestock population, largely due to the limited employment opportunities in other economic sectors[3].

Cattle have played a key role in human history, having been domesticated since at least the early Neolithic age. Archaeozoological and genetic data indicate that cattle were first domesticated from wild aurochs (Bos primigenius) approximately 10,500 years ago. There were two major areas of domestication: one in central Anatolia, the Levant, and Western Iran, which gave rise to the taurine line, and a second in the area that is now Pakistan, which produced the indicine line[4]. Since this divergence, the genomes of Bos taurus and Bos indicus cattle have undergone mutations and selection pressure, ultimately enabling cattle to become better adapted to their environments[5]. As domestication spread across regions, cattle adapted to different ecological niches and ultimately diverged into two major subspecies: Bos taurus, which adapted to temperate and cooler climates and expanded across Europe and West–Central Africa, and Bos indicus, which evolved to thrive in hot, humid tropical regions and became widely distributed across North Africa, China, and the Indian subcontinent[4].

India hosts 53 officially recognized native Bos indicus breeds, each uniquely adapted to the nation's varied tropical and subtropical environments. Breeds such as Gir, Sahiwal, Red Sindhi, and Kankrej have evolved over centuries to withstand heat

stress, resist diseases and parasites, and support the livelihoods of millions of rural farmers, forming an essential part of India's agrarian economy[6]. While Bos indicus has many features that make it well-adapted to the harsher, hot, humid climate, this advantage comes at the expense of lower milk output per cow. To address this challenge, a few European Bos taurus breeds were introduced into tropical regions and have since undergone selection through crossbreeding with native breeds.

Categorizing this diversity of cattle breeds is important for multiple reasons. Many breeds exhibit distinct traits, such as milk fat content, meat quality, disease resistance, and environmental adaptability, making them valuable for specific agricultural and commercial purposes[7].

Traditional approaches to breed classification primarily rely on phenotypic identification or limited genetic marker panels. However, phenotypic methods are subjective and prone to misclassification, whereas low-density marker panels often lack sufficient resolution. Pedigree-based identification offers another alternative, but reliable breeding records are largely unavailable in the Indian context, where cattle ownership is predominantly distributed among small-scale farmers[8].

Advances in genomics have enabled breed identification using high-density genotyping arrays (e.g., Illumina BovineSNP50 BeadChip) and whole-genome sequencing. Although these approaches provide a high-resolution view of genetic variation, their cost and infrastructure requirements limit their scalability, particularly for routine breed identification or large-scale commercial validation.

Machine learning (ML) has emerged as a powerful framework for analyzing high-dimensional biological data. In genomics, ML models can learn complex patterns from genome-wide variants, such as single-nucleotide polymorphisms (SNPs), and have been shown to accurately classify individuals into predefined categories. Importantly, ML-based approaches also enable feature selection, allowing the identification of a minimal subset of informative genetic markers. This creates an opportunity to develop cost-effective and scalable solutions for breed classification by reducing the number of required SNPs without compromising predictive performance[9, 10].

Chapter 2

Related Research

Cattle breed identification has shifted away from traditional, error-prone approaches to genetic characterization due to their higher resolution. Genetic characterization and breed identification in cattle rely on genome-wide single-nucleotide polymorphism (SNP) markers due to their high abundance, genome-wide distribution, and reproducibility. SNP-based approaches are now widely used for breed assignment, population structure analysis, and genomic selection. In the context of indigenous cattle, especially in India, SNP-based studies have been extensively applied to characterize genetic diversity and identify breed-specific variations across diverse populations[11, 12].

A major line of research focuses on identifying breed-informative SNP panels. Several studies have applied statistical measures, such as the fixation index (F_{ST}) and allele frequency differences, and multivariate approaches, such as principal component analysis, to identify discriminative SNPs. These panels are designed to reduce dimensionality while retaining classification power. For example, studies have demonstrated that optimized SNP subsets can effectively distinguish cattle breeds and can be used for traceability and conservation purposes [6]. Similarly, recent work has emphasized the utility of SNP panel reduction strategies to maintain high classification accuracy with fewer markers, improving computational efficiency and applicability[13, 14].

In the Indian context, efforts have been made to identify breed-specific SNPs for indigenous cattle populations. These studies typically involve extensive filtering and comparative genomic analysis to derive informative marker sets [3,4]. While effective, such approaches are often computationally intensive and highly dependent on dataset-specific characteristics, limiting their scalability across multiple breeds.

More recently, machine learning (ML) approaches have been introduced for cattle breed classification using genomic data. Algorithms such as support vector machines

(SVMs), random forests (RFs), k-nearest neighbors (KNNs), and neural networks have been applied to SNP datasets to improve classification accuracy[15, 16]. These methods demonstrate strong predictive performance, especially when combined with optimized SNP panels. Additionally, ML-based feature selection techniques have been shown to outperform traditional statistical filtering in identifying informative markers[14].

However, despite these advancements, existing ML-based genomic studies exhibit several limitations. A key issue is that most models are developed using datasets from European or Chinese cattle populations, which predominantly represent taurine breeds or well-characterized commercial lines[17, 15]. These populations differ significantly from indigenous Indian (indicine) breeds in terms of genetic structure, adaptation, and diversity, limiting the transferability of such models. Furthermore, while SNP discovery studies in Indian cattle successfully identify breed-informative markers, they are largely marker-centric rather than model-centric. That is, they focus on identifying SNPs but do not systematically develop or evaluate robust machine learning frameworks for classification across multiple breeds. Even when ML approaches are used, they are often limited to specific datasets or a small number of breeds, restricting generalization[14].

Another important limitation is the lack of region-specific ML models that account for the genetic diversity and environmental variability characteristic of Indian livestock systems. Studies show that genomic prediction and classification models often exhibit reduced robustness when applied across genetically distinct populations or under different environmental conditions[17]. A further limitation of existing approaches is the lack of interpretability. Many studies prioritize predictive performance without systematically evaluating the contribution of individual SNPs or genomic regions to classification. This restricts biological insight, as the genetic determinants underlying breed differentiation and divergence are not adequately characterized[11].

Overall, the literature indicates that while breed-specific SNP identification is well established, and machine learning models for breed classification exist, there is a lack of integration between these approaches for Indian cattle populations.

2.1 Research Gap

From the reviewed literature, the following gaps can be identified:

- Breed-specific SNP panels exist for Indian cattle but are labor-intensive, dataset-dependent, and not easily scalable.

- Machine learning models for genomic breed classification are largely developed on European or Chinese datasets, limiting applicability to Indian breeds.
- There is a lack of integrated, generalizable ML frameworks trained on Indian cattle SNP data for robust breed classification. This is necessary as there is a poor generalization across populations.
- Indian cow breeds are difficult to classify as some breeds are highly similar.
- Most ML models for breed identification have small sample size and generally no measures are taken to reduce overfitting.

To address these research gaps, this study uses whole-genome variant and genotype data from multiple Indian cow breeds to develop a machine learning model that accurately classifies individuals into their respective breeds.

Chapter 3

Methodology

3.1 Datasets

This project used two genomic datasets:

1. A whole genome sequence variant VCF dataset from Indian cow breeds with 120 samples of 9 breeds and 39.5 million variants.

2. A publicly available array-genotype PED dataset from WIDDE[18] with 224 samples from 6 breeds with 45.9k variants. The sequences were obtained using either the Illumina BovineHD[19, 20] or Decker et al[21].

And an image dataset: Indian Bovine breed image dataset from Kaggle[22]. The images were selected only for the breeds whose genomic data were used in this project. Some breeds did not have any images. The table below describes the samples across breeds.

Breed	**No. of Genomic Samples**	**No. of Images**
Brahman	79	-
Nelore	60	-
Gir	59	372
Holstein	46	328
Sahiwal	32	439
Hereford	22	-
Ongole	20	191
Jersey	13	203
Tharpakar	13	217

Table 3.1: Breed-wise distribution of genomic samples and images

3.2 Genomic Data Processing

3.2.1 Remapping and formatting

The WIDDE PED data were converted to BED format using PLINK 1.9[23, 24]. The WGS data originally were aligned to the ARS-UCD1.2 genome, while the WIDDE data were aligned to the older UMD3.1 genome. To make them comparable, the WIDDE data were lifted from bosTau6 to bosTau9 coordinates using the UCSC LiftOver tool[25] with default settings. After liftover, the coordinates in the WIDDE BED file were updated. Sample names were also adjusted: column 1 contains the breed, and column 2 contains breed + number. SNP IDs were reformatted to "chr: position" using PLINK2[26].

3.2.2 Preprocessing and merging

The WGS and updated WIDDE datasets were merged using PLINK 1.9. Only SNPs present in both datasets were kept. The WIDDE dataset was strand-flipped to match the WGS data. The merged dataset contained 344 samples and 36.3k variants.

3.2.3 Quality check (QC) and filtering

The merged file underwent QC and preprocessing for the following:

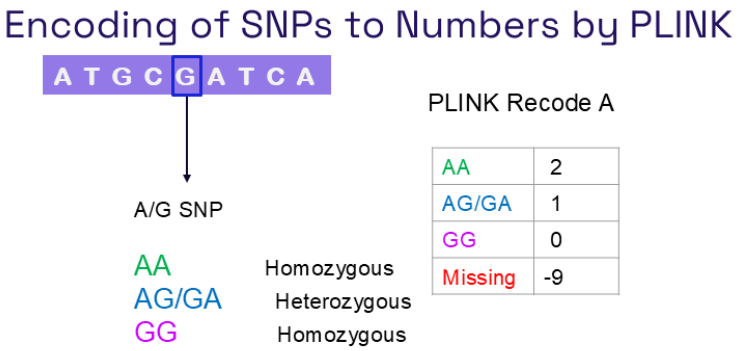
- **Genotyping rate**: Indicates the percentage of successfully called genotypes, generally calculated as total non-missing calls divided by total possible calls. It is a critical Quality Control (QC) metric used to remove low-quality SNPs[27].
- **Minor allele frequency**: The frequency at which the second most common allele (the rare variant) occurs in a given population[28].
- **Linkage Disequilibrium**: the non-random association of alleles at different loci within a population, meaning certain alleles are inherited together more often than expected by chance[29].

The merged data were filtered using PLINK based on the following criteria:
Genotyping rate: $\geq 95\%$
Minor allele frequency: $\geq 5\%$
Linkage Disequilibrium: ≥ 0.5 (with window size 50 and step size 50)

After filtering, the dataset contained 344 samples and 11,118 variants.

3.2.4 Numerical Encoding

The final dataset was converted to a numeric format using the PLINK `recodeA` command to facilitate the downstream machine learning processing. The encoding works as follows:



The VGG-16 model is a convolutional neural network (CNN) architecture that was proposed by the Visual Geometry Group (VGG) at the University of Oxford. It is characterized by its depth, consisting of 16 layers including 13 convolutional layers and 3 fully connected layers.

Figure 3.1: Numerical encoding in PLINK

The resulting BED file was converted to RAW format to combine the information from the BED, FAM, and BIM files into one.

3.3 Machine Learning for Genomic Data

3.3.1 Overview

- Input data (X): High-dimensional genotype matrix (344 samples x 11118 variants)
- Output (y): Breed labels (9 categorical classes)
- Task: Supervised multi-class classification
- Objective: Maximize classification performance (Accuracy and ROC-AUC) and identify the genomic features contributing to classification.

3.3.2 Dimensionality Reduction

Dimensionality reduction was performed using Uniform Manifold Approximation and Projection (UMAP)[30], a non-linear method that captures both local and global

structure in high-dimensional data. Unlike Principal Component Analysis (PCA), which is limited to linear relationships, UMAP can model more complex patterns commonly found in biological datasets. Compared to t-Distributed Stochastic Neighbor Embedding (t-SNE), UMAP is generally faster, preserves more of the overall data structure, and produces more consistent results across runs.

UMAP was applied using three distance metrics: Euclidean, Cosine, and Hamming to capture different types of similarity in the data. The parameters were set to n_neighbors = 15, which balances local and global structure, and min_dist = 0.1, which allows clear clustering while maintaining some separation between groups.

3.3.3 Data split

The data was split into training (80%) and test (20%) sets using scikit-learn's `train-test-split`.

3.3.4 Feature Selection

To select only the relevant features for model training, the following feature selection methods were applied on the training set:

- **Mutual info gain (MIG)**: Mutual information, from Information Theory, measures how much knowing one variable reduces uncertainty about another. In feature selection, it quantifies how informative a feature X is with respect to the target Y[31].
- **Recursive feature elimination (RFE)**: Recursive Feature Elimination is a feature selection method that iteratively trains a model, ranks features by importance, and removes the least important ones until a desired subset is obtained[32]. For linear models, feature importance is typically based on the magnitude of coefficients. Features with the smallest importance are removed at each iteration.

The top 1000 SNPs were selected using both methods for model development. The same 1000 SNPs were extracted from the test set for model evaluation.

3.3.5 ML Models

The following supervised models were trained in two ways: one using MIG SNPs and another using RFE SNPs with 10-fold cross-validation to prevent overfitting.

- **Logistic Regression**: models the probability of a binary outcome using a logistic (sigmoid) function applied to a linear combination of features.

- **Random Forest Classifier**: an ensemble method that builds multiple decision trees on bootstrapped subsets of data and aggregates their predictions (majority voting for classification).

- **Support Vector Machine**: finds a hyperplane that maximizes the margin between classes. With kernels, it can model non-linear boundaries.

- **k-nearest Neighbors**: a non-parametric method that classifies a sample based on the majority label among its k nearest neighbors in feature space.

3.3.6 Final model training

Logistic Regression was selected as the final model for training based on preliminary results. The data was split into three sets to use the validation set for hyperparameter fine-tuning. Training (70%), validation (10%), and test (20%) sets were split using scikit-learn's `train-test-split` with two stratified splits: 70% & 30%, then 30% into 2/3 and 1/3.

To improve performance, feature selection was performed using L1 Regularization.

- **L1 Regularization**: Technique used to prevent overfitting by penalizing the size of model coefficients during training. It encourages the model to keep only the most important features by shrinking less important ones exactly to zero. As a result, L1 regularization not only improves generalization but also performs automatic feature selection, making the model simpler and more interpretable.

- **Optuna hyperparameter tuning**: Optuna is a tool for automatic hyperparameter tuning. Rather than random guessing, Optuna uses advanced algorithms to find the best hyperparameters. For the Logistic Regression with L1 Regularization model, C-value is a hyperparameter that controls the strength of the regularization effect with smaller values specifying stronger regularization.

  Optuna was used with 50 trials with C-value range from 10^-4 to 10. It used the training set for model training and the validation set for C-value tuning. The obejctive was to maximize the AUC. The test set was kept untouched.

3.4 Deep Learning for Cattle Image Classification

A VGG16 model was used to classify the image dataset. Image augmentation of Horizontal flip, Rotation, Zoom, Height and Width changes were applied as the number of images was low. A VGG16 model was used for image classification.

- VGG16: The VGG16 model is a convolutional neural network (CNN) architecture that was proposed by the Visual Geometry Group (VGG) at the University of Oxford. It is characterized by its depth, consisting of 16 layers including 13 convolutional layers and 3 fully connected layers.

The model is based on the pre-trained VGG16 architecture using transfer learning. The convolutional base of VGG16 (trained on ImageNet) is used as a fixed feature extractor, and custom layers are added for classification.

The input images are resized to $150 \times 150 \times 3$. Data augmentation is applied during training to improve generalization.

The extracted features from VGG16 are passed through a Global Average Pooling layer, followed by two fully connected layers with 512 and 256 neurons respectively, both using ReLU activation and Dropout (0.5) for regularization.

The final layer is a Dense layer with 6 neurons and softmax activation, which outputs probabilities for the 6 classes.

Training stopped early at 28 epochs due to `EarlyStopping`. The validation loss did not improve for 8 consecutive epochs, so training was automatically terminated to prevent overfitting. The best model weights were restored.

The model architecture is as follows:

Layer (type)	**Output Shape**	**Param**
input_layer_1 (InputLayer)	(None, 150, 150, 3)	0
data_augmentation (Sequential)	(None, None, None, 3)	0
vgg16 (Functional)	(None, None, None, 512)	14,714,688
global_average_pooling2d (GlobalAveragePooling2D)	(None, 512)	0
dense (Dense)	(None, 512)	262,656
dropout (Dropout)	(None, 512)	0
dense_1 (Dense)	(None, 256)	131,328
dropout_1 (Dropout)	(None, 256)	0
dense_2 (Dense)	(None, 6)	1,542
Total params		15,110,214 (57.64 MB)
Trainable params		395,526 (1.51 MB)
Non-trainable params		14,714,688 (56.13 MB)

Table 3.2: Model Architecture Summary

Chapter 4

Results

4.1 UMAP

The UMAP plots of the data with three metrics are as follows:

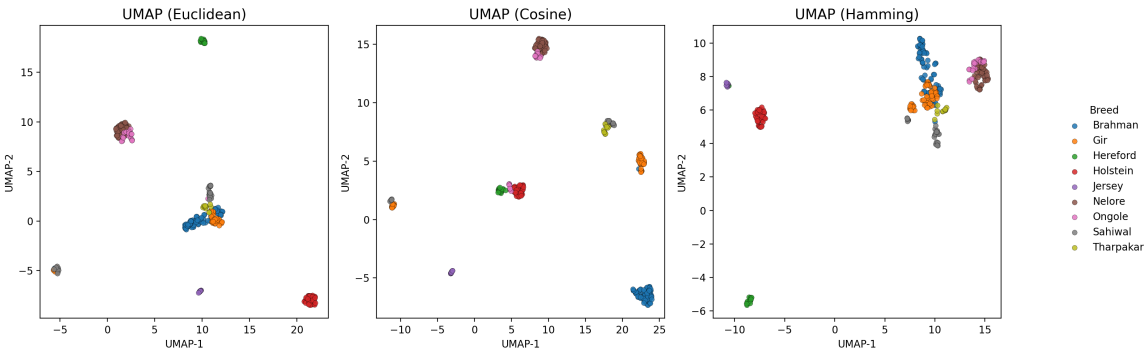
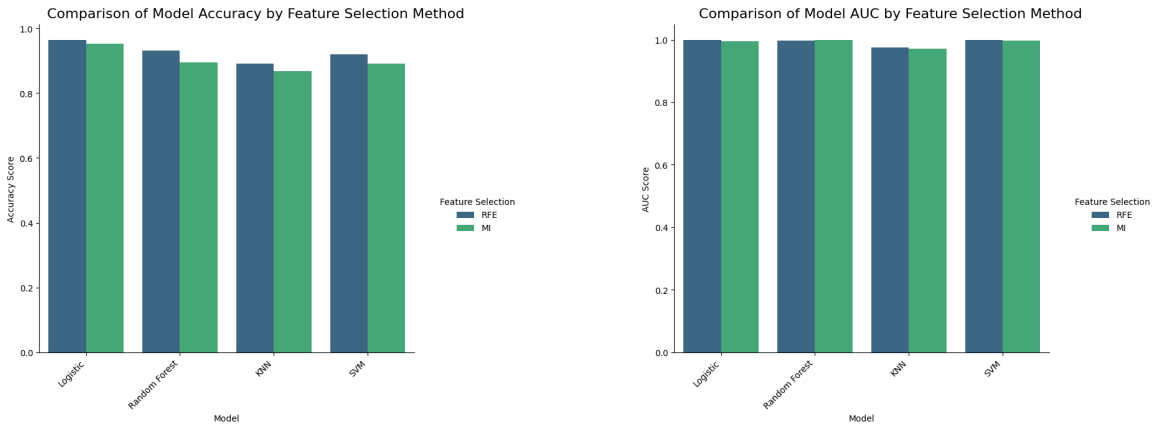


Figure 4.1: UMAP of the dataset

4.2 Preliminary ML models

The accuracy and AUC of the preliminary model training and evaluation for the individual models are as follows:



(a) AUC (b) Accuracy

Figure 4.2: Performance of ML models

The combined metrics of the models are summarized in the table below. The confusion matrices and ROC-AUC plots of individual models are in the appendix A.

4.3 Logistic + L1 model

4.3.1 C value tuning using Optuna

The results for tuning of of C-value are as follows:

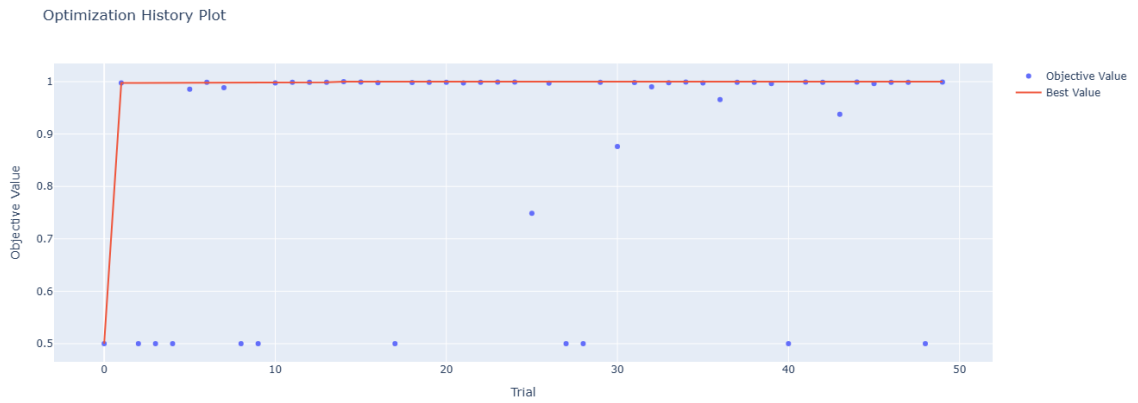
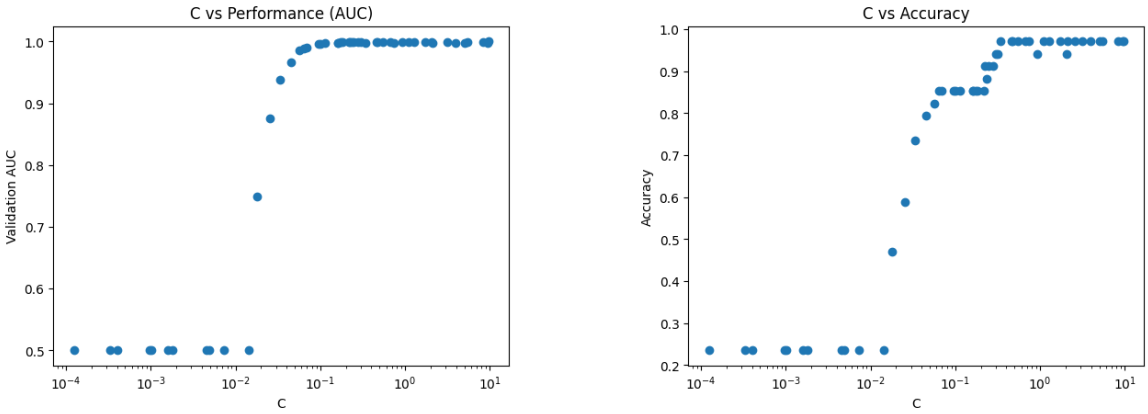


Figure 4.3: Optuna optimization plot

The C-value vs Accuracy and AUC charts are as follows:



(a) AUC (b) Accuracy

Figure 4.4: C-value vs model performance

4.3.2 10 fold Cross-validation

The 10-fold Cross validation combined metrics are as follows:

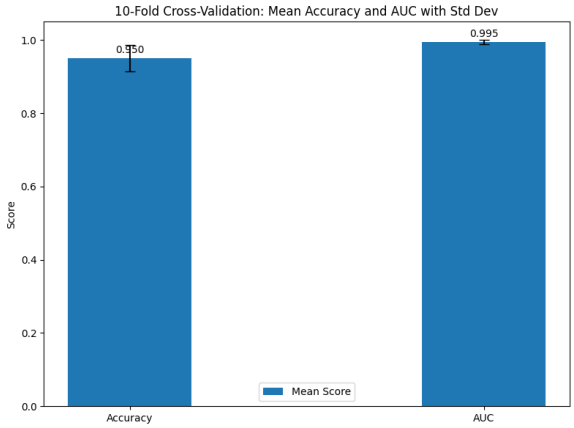


Figure 4.5: Mean Accuracy and AUC of 10-fold CV

The per-fold Accuracies and AUCs are as follows:

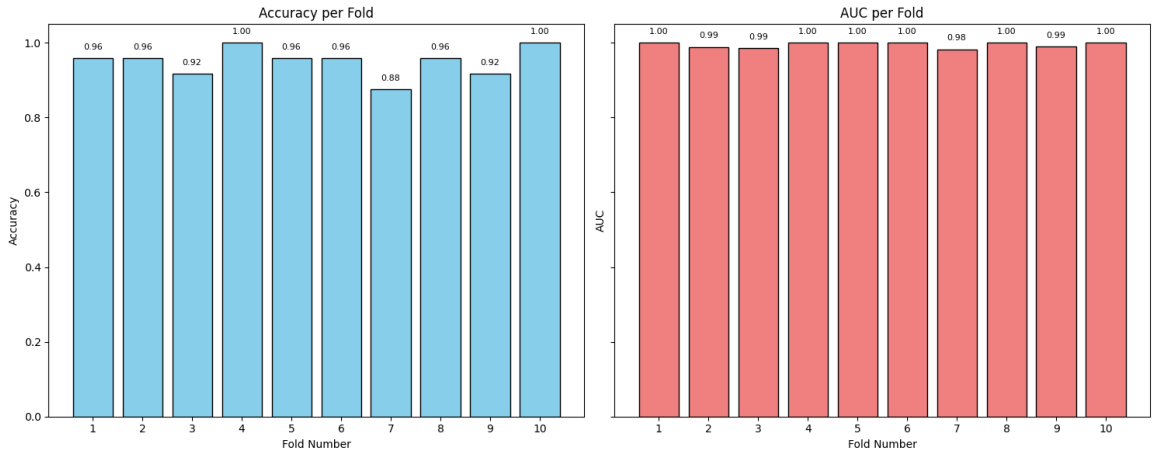
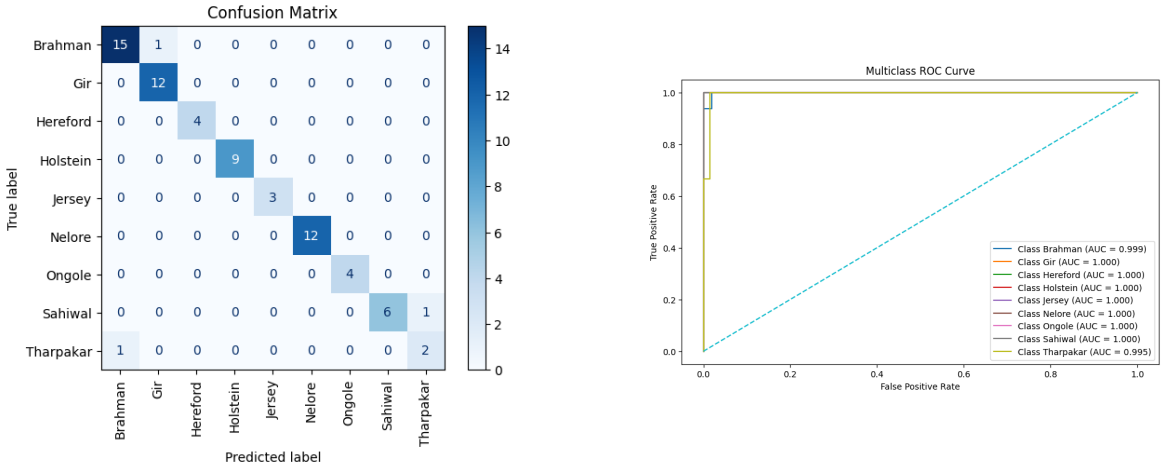


Figure 4.6: Per-fold model 10-fold CV metrics

4.3.3 Final model performance

The confusion matrix and ROC-AUC curves for the final model on test data is as follows:



(a) Confusion Matrix (b) ROC-AUC curves

Figure 4.7: Final model performance

The classification report of the model is as follows:

Breed	Precision	Recall	F1-score	Support
Brahman	0.94	0.94	0.94	16
Gir	0.92	1.00	0.96	12
Hereford	1.00	1.00	1.00	4
Holstein	1.00	1.00	1.00	9
Jersey	1.00	1.00	1.00	3
Nelore	1.00	1.00	1.00	12
Ongole	1.00	1.00	1.00	4
Sahiwal	1.00	0.86	0.92	7
Tharpakar	0.67	0.67	0.67	3
Accuracy			0.96	70
Macro Avg	0.95	0.94	0.94	70
Weighted Avg	0.96	0.96	0.96	70

Table 4.1: Performance metrics of the model on test

4.3.4 Biological Features Selected

The L1 regularization selected **816 SNPs** from the 29 chromosomes.

The plot of total features to selected features for each bovine chromosome is as follows:

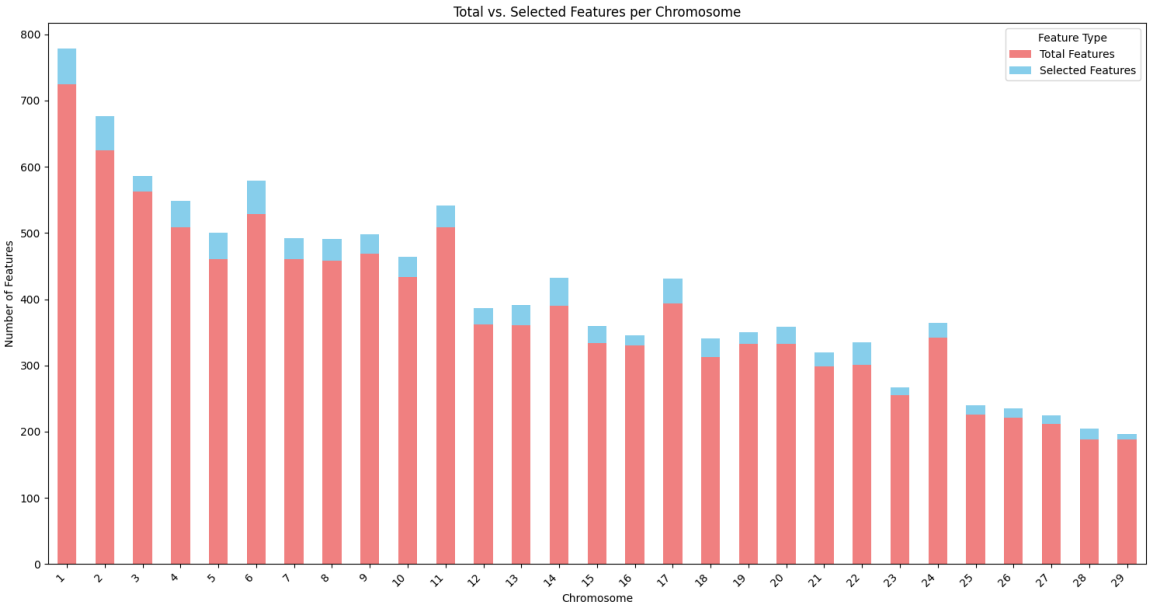


Figure 4.8: Selected Features per chromosome

The chromosomes with top and least numbers of features selected are given in tables below.

Chromosome	Total Features	Selected Features	Percentage Selected (%)
22	301	34	11.30
14	390	42	10.77
6	528	51	9.66
17	394	37	9.39
18	312	29	9.29

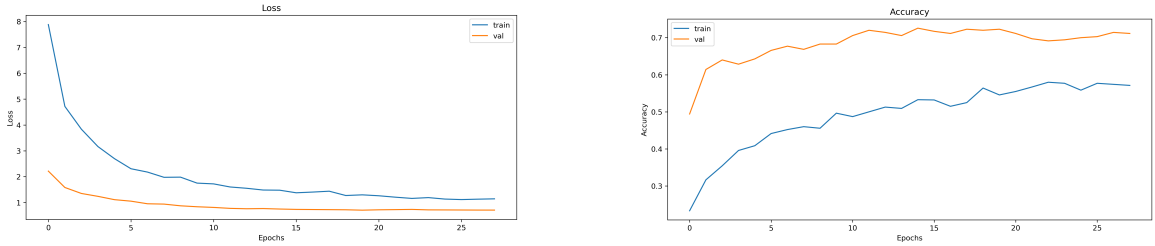
Table 4.2: Top 5 chromosomes with highest percentage of selected features

Chromosome	Total Features	Selected Features	Percentage Selected (%)
29	188	8	4.26
3	562	24	4.27
23	255	12	4.71
16	330	16	4.85
19	333	17	5.11

Table 4.3: Bottom 5 chromosomes with lowest percentage of selected features

4.4 VGG16 model

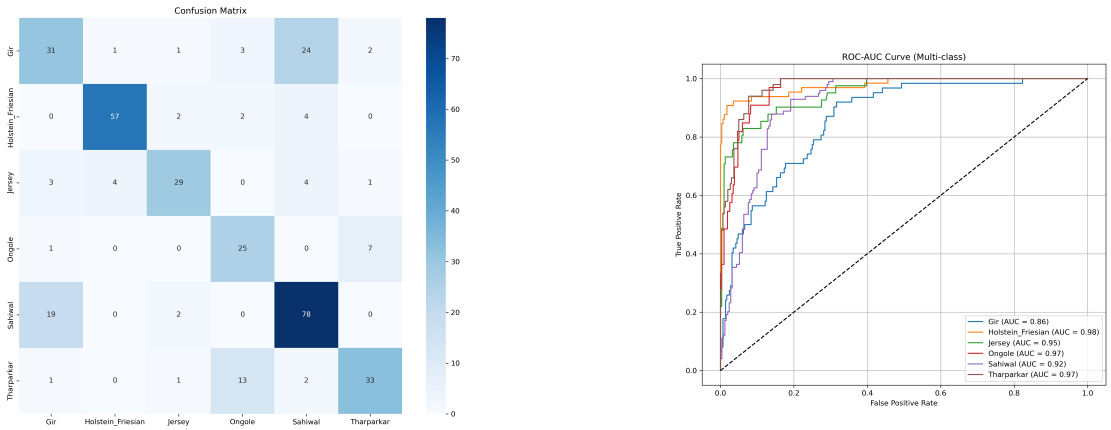
The loss and accuracy curves for training and validation are as follows:



(a) Loss (b) Accuracy

Figure 4.9: VGG16 model

The multi-class confusion matrix and ROC-AUC curves are as follows:



(a) Confusion Matrix (b) ROC-AUC curve

Figure 4.10: VGG16 model

The classification report is as follows:

Breed	**Precision**	**Recall**	**F1-score**	**Support**
Gir	0.56	0.5	0.53	62
Holstein Friesian	0.92	0.88	0.9	65
Jersey	0.83	0.71	0.76	41
Ongole	0.58	0.76	0.66	33
Sahiwal	0.7	0.79	0.74	99
Tharpakar	0.77	0.66	0.71	50
Accuracy			0.72	350
Macro Avg	0.73	0.71	0.72	350
Weighted Avg	0.73	0.72	0.72	350

Table 4.4: Performance metrics of VGG16 model

Chapter 5

Discussion

The results demonstrate that Machine Learning methods applied to SNP-based classification can achieve high accuracy in distinguishing Indian cow breeds. The Logistic regression model with L1 Regularization achieved an AUC of 0.999, meaning that the model is highly efficient at distinguishing one breed from another. The strong performance obtained with just 816 SNPs suggests that the subset of markers selected via L1 regularization effectively captures the most informative breed-specific genetic variation, enabling accurate classification despite substantial marker reduction.

The reduction in the number of SNPs to fewer than 1000 represents a major advantage of the proposed model, as it substantially lowers genotyping costs while maintaining high classification accuracy. This cost-effectiveness enhances its practical applicability across multiple domains, including breed identification in local cattle breeding programs, the development of breed registries and conservation initiatives, and commercial settings where it can be used to verify claims regarding products sourced from specific cattle breeds.

Overall, these findings suggest that machine learning–based methods provide a more accurate and cost-efficient alternative to conventional breed identification techniques. By overcoming the low resolution associated with phenotypic observation and microsatellite markers, while requiring fewer features than approaches based on whole-genome sequencing or high-density SNP genotyping, this approach offers an efficient and scalable solution for breed classification.

Despite these promising results, several limitations must be considered. The relatively small sample size (n = 344) may limit the generalizability of the model to broader populations. In addition, the use of merged SNP datasets containing only markers common to both SNP chip and WGS platforms may have resulted in the exclusion of potentially informative variants unique to whole-genome data. Although

simpler models such as logistic regression reduce the risk of overfitting and improve interpretability, the extremely high AUC observed in this study may still indicate potential overfitting, particularly given the limited dataset.

Furthermore, class imbalance across breeds, with some breeds being overrepresented, may have introduced bias into the model. The study is also limited to only nine breeds, whereas India has a much larger diversity of cow breeds with just the number of officially recognized breeds being 51. However, the availability of publicly accessible WGS and SNP chip data for Indian cow breeds is limited, making it difficult to expand the sample size and groups.

In addition to genomic classification, the VGG16 model used for image-based breed classification in this study is preliminary and could be further improved by incorporating a larger and more diverse image dataset, as well as exploring alternative deep learning architectures.

Similarly, the logistic regression model could be improved in the future with a larger sample size and could incorporate a higher number of breeds. The development of a targeted SNP panel consisting of a minimal set of highly informative markers could further enhance the cost-effectiveness and practical deployment of this approach.

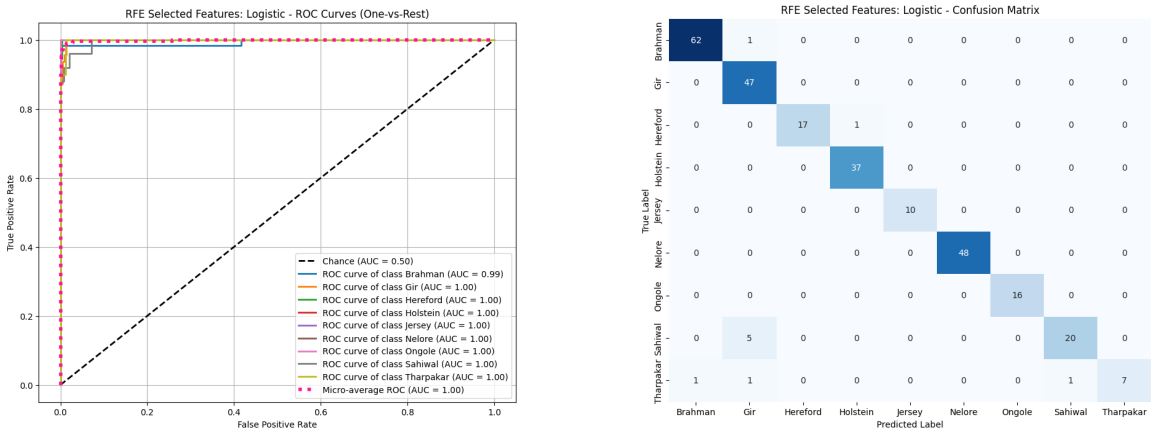
Additionally, a comprehensive breed identification system can be developed by integrating both image-based and genomic-based models. This multimodal approach has the potential to improve overall classification performance, as certain breeds may be poorly distinguished by one modality alone. For example, some breeds may show lower accuracy in genomic data, while others may be harder to classify using only image features. By combining both data sources, the strengths of one model can compensate for the weaknesses of the other, leading to more robust and accurate breed identification.

Appendix A

Supplementary Results

A.1 Preliminary ML model results

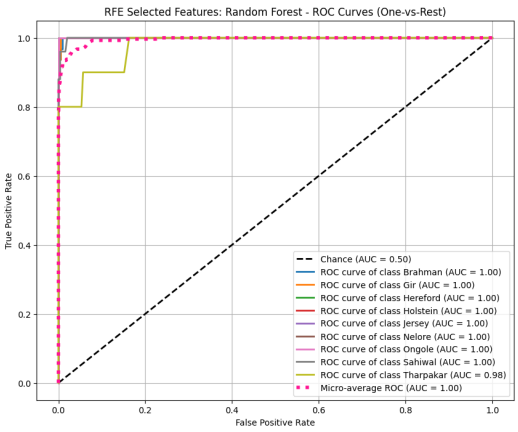
A.1.1 Recursive Feature Elimination



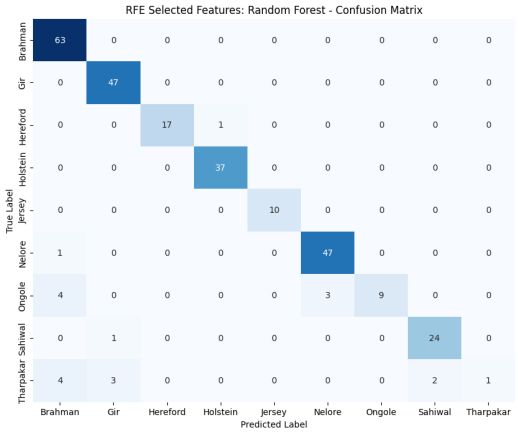
(a) ROC-AUC

(b) Confusion Matrix

Figure A.1: Logistic Regression

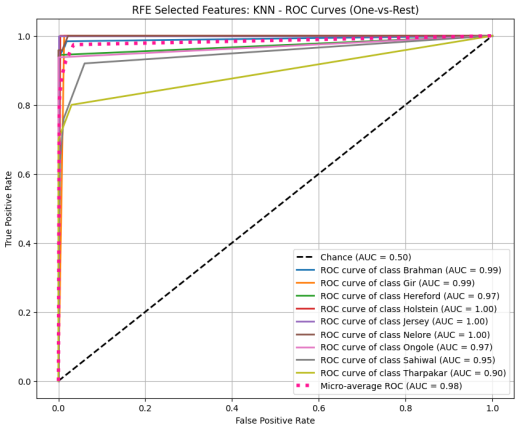


(a) ROC-AUC

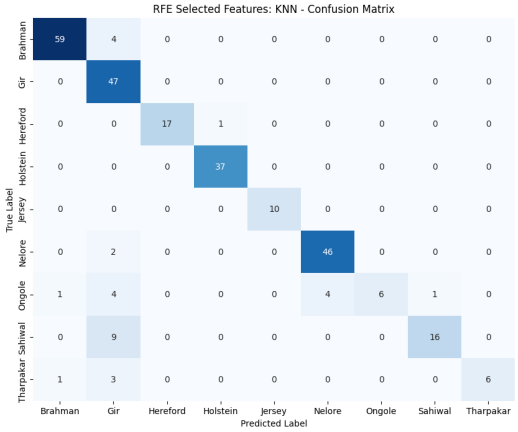


(b) Confusion Matrix

Figure A.2: Random Forest Classifier

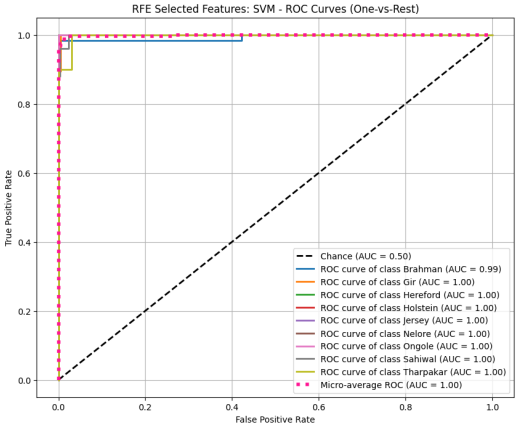


(a) ROC-AUC

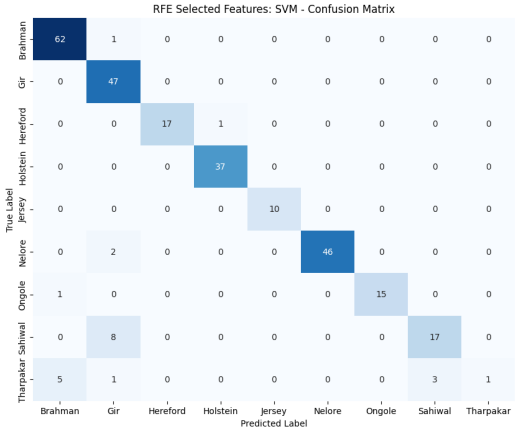


(b) Confusion Matrix

Figure A.3: k-Nearest Neighbors



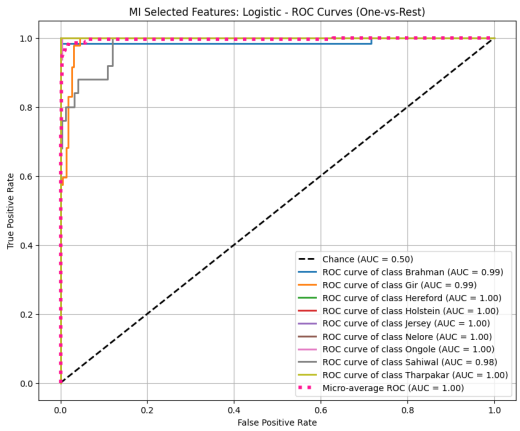
(a) ROC-AUC



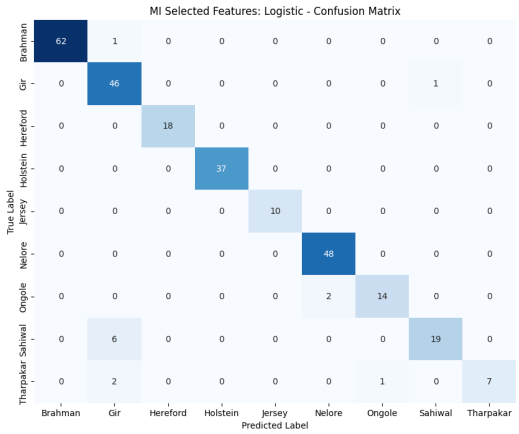
(b) Confusion Matrix

Figure A.4: Support vector Machine

A.1.2 Mutual Info gain

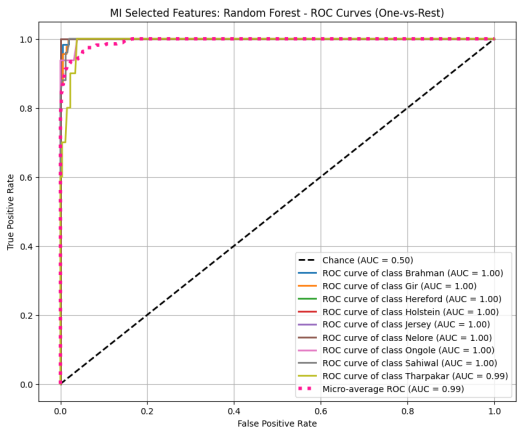


(a) ROC-AUC

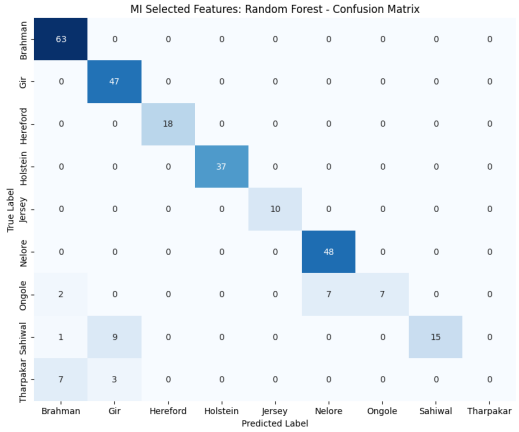


(b) Confusion Matrix

Figure A.5: Logistic Regression

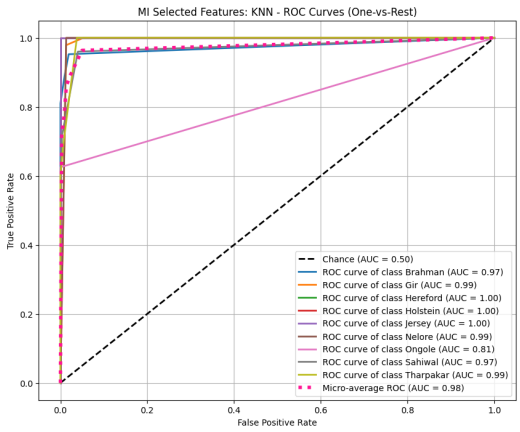


(a) ROC-AUC

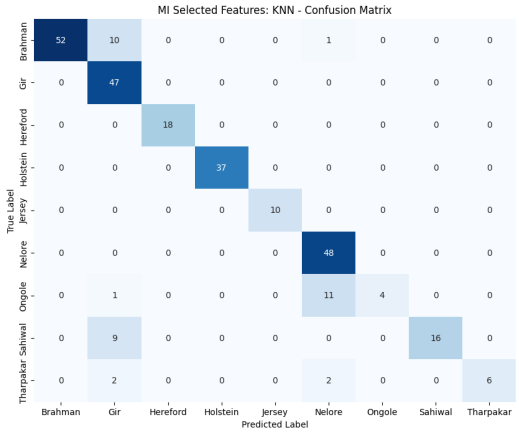


(b) Confusion Matrix

Figure A.6: Random Forest Classifier

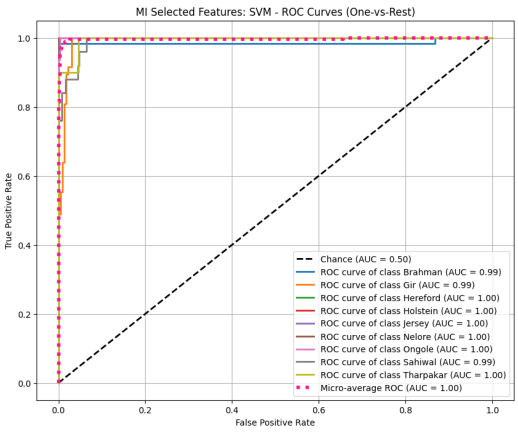


(a) ROC-AUC

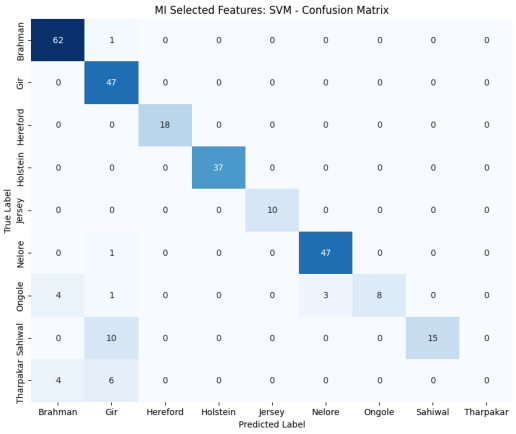


(b) Confusion Matrix

Figure A.7: k-Nearest Neighbors



(a) ROC-AUC



(b) Confusion Matrix

Figure A.8: Support vector Machine

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